



MUTHAYAMMAL ENGINEERING COLLEGE

(Autonomous)

Rasipuram – 637 408, Namakkal (Dt.), Tamil Nadu, India.

Department of Artificial Intelligence and Data Science

LAB MANUAL

23ADC47 / DATA MODELLING AND BUSINESS INTELLIGENCE LABORATORY

Prepared By

Mrs.R.K.Nithya, AP/AI&DS

Ms.P.Kavya, AP/AI&DS

Lab Handling Faculty : _____

Year / Sem /Sec : _____

Lab Incharge

Course Coordinator

HoD

Principal



MUTHAYAMMAL ENGINEERING COLLEGE

(Autonomous)

Rasipuram - 637 408, Namakkal (Dt.), Tamil Nadu, India.

Institute Vision & Mission

Institution Vision

- To be a Centre of Excellence in Engineering, Technology and Management on par with International Standards.

Institution Mission

- To prepare the students with high professional skills and ethical values.
- To impart knowledge through best practices.
- To instill a spirit of innovation through Training, Research and Development.
- To undertake continuous assessment and remedial measures.
- To achieve academic excellence through intellectual, emotional and social stimulation.

Department Vision & Mission

Department Vision

- To create an inspirational learning centre where proficient and future-ready scientist in the field of Artificial Intelligence and Data Science.

Department Mission

- To impart high-quality education and capitalist oriented learning through Artificial Intelligence and Data Science.
- To contribute towards advanced AI technologies that provides increased and better performance.
- To benefit the society through our contribution towards advancements in AI and Data Science.

Program Educational Objectives

- PE01** : Graduate should be able to pursue as an Engineer with necessary conceptual, analytical and theoretical knowledge in the domain of Electronics and Communication Engineering.
- PE02** : Graduate should be able to acquire the practical knowledge through basics and advanced laboratories in the field of Electronics and Communication Engineering.
- PE03** : Graduate should be able to demonstrate the leadership skills through Entrepreneurship, Employment and Higher Studies and to practice ethical values for the benefit of Society and Environment.

Program Outcomes

- P01** : **Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- P02** : **Problem Analysis:** Identify, formulate, review research literature and analyse complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).
- P03** : **Design/Development of solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).
- P04** : **Conduct investigations of complex problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- P05** : **Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).
- P06** : **The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
- P07** : **Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).
- P08** : **Individual and Collaborative Team Work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

- P09 : Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
- P010 : Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- P011 : Life-long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).

Program Specific Outcomes

- PS01** : Design and develop electronic circuits and systems using suitable hardware and software tools for diverse applications.
- PS02** : Apply acquired knowledge and analytical skills for modeling, simulation, and implementation of advanced communication systems.

23ADC47

Data Modeling And Business Intelligence Laboratory

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Course Objective:

- To gain the ability to design and maintain efficient
- To scalable business intelligence solutions
- Sift data using slicers in multiple pivot tables
- Students will execute statistical analyses with professional statistical software.

Course Outcomes:

- 23ADC47.CO1 Be able to analyze large data sets to support data-driven decision making.
- 23ADC47.CO2 understanding of machine learning and statistical analysis techniques
- 23ADC47.CO3 able to discriminate between descriptive, diagnostic, predictive, and prescriptive analytics
- 23ADC47.CO4 develop the ability to build and assess data-based models.
- 23ADC47.CO5 execute statistical analyses with professional statistical software

Course Outcomes	Program Outcomes												Program Specific Outcomes		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
23ADC47.CO1	X	X	X	x	x	x	X	x	x	x	x	x	x	x	x
23ADC47.CO2	X	X	X	x	x	x	X	x	x	x	x	x	x	x	x
23ADC47.CO3	X	X	X	x	x	x	X	x	x	x	x	x	x	x	x
23ADC47.CO4	X	X	X	x	x	x	x	x	x	x	x	x	x	x	x
23ADC47.CO5	X	X	X	x	x	x	x	x	x	x	x	x	x	x	x

SL.No.

List of Experiments

1. Installation and Configuration of tableau and airflow
2. Collecting, Cleaning and Connecting to data.
3. Perform ETL process for the given data source.
4. Create charts like bubble, bar, map using Airflow and Tableau.
5. Create your story from the charts with valid reasons.
6. Build a Map view using Tableau (Explore your data geographically)
7. Collect appropriate data, Perform ETL process and develop Scorecard using Air flow and Tableau
8. Collect appropriate data, Perform ETL process and develop dashboard for Health Care using Air flow and Tableau

Total Periods : 60


Chairman
Board of Studies
 Department of Computer Science and Engineering
MUTHAYAMMAL ENGINEERING COLLEGE
 (AUTONOMOUS)
 RASIPURAM-637400, NAMAKKAL Dt.,
 TAMIL NADU

INDEX

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7.	Collect appropriate data, Perform ETL process and develop Scorecard using Air flow and Tableau.
8.	Collect appropriate data, Perform ETL process and develop dashboard for Health Care using Air flow and Tableau.

Ex. No. : 1	Installation and Configuration of tableau and airflow.
Date :	

AIM

To install and study about the features in tableau desktop version.

ALGORITHM**BASICS OF TABLEAU**

- Tableau was founded in 2003 as a result of a computer science project at Stanford that aimed to improve the flow of analysis and make data more accessible to people through visualization.
- Co-founders Chris Stolte, Pat Hanrahan, and Christian Chabot developed and patented Tableau's foundational technology, VizQL—which visually expresses data by translating drag-and-drop actions into data queries through an intuitive interface
- Tableau is a leading data visualization tool used for data analysis and business intelligence.
- Tableau is an excellent data visualization and business intelligence tool used for reporting and analyzing vast volumes of data.
- It helps users create different charts, graphs, maps, dashboards, and stories for visualizing and analyzing data, to help in making business decisions.
- Tableau has a lot of unique, exciting features that make it one of the most popular tools in business intelligence (BI).

TABLEAU FEATURES

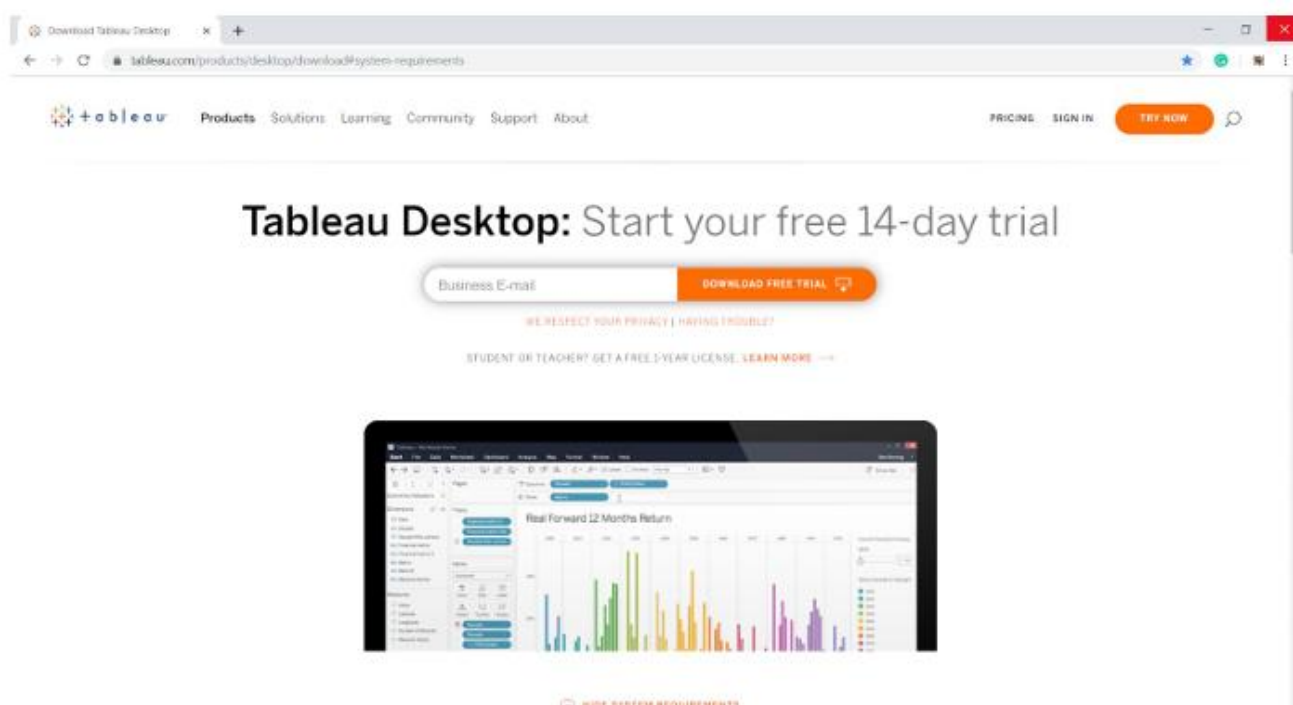
- Tableau supports powerful data discovery and exploration that enables users to answer important questions in seconds
- No prior programming knowledge is needed; users without relevant experience can start immediately with creating visualizations using Tableau
- It can connect to several data sources that other BI tools do not support. Tableau enables users to create reports by joining and blending different datasets
- Tableau Server supports a centralized location to manage all published data sources within an organization

PROCEDURE :

DOWNLOADING STEPS

Step 1: To download Tableau Desktop, first we need to open our browser and go <https://public.tableau.com/en-us/s/download> . Once the website opens, click on the TRY NOW button to initiate the download process.

Step 2: This will take us to the download page for Tableau Desktop. When we scroll down on this page, we will get a list of system requirements to download Tableau Desktop. Here, they have provided platform wise requirements, that is, for Windows, Mac, etc. Make sure to check the requirements for your platform before starting the download.



Step 3: At the top of this page, it will ask you to enter a business E-mail. Please note that you must not enter your personal E-mail ID here. Enter the appropriate E-mail address and click on DOWNLOAD FREE TRIAL.

Step 4: As soon as you will click on the download button, a .exe file will start downloading automatically. You will find a file download icon at the left-most bottom of your screen. It will take around 4 to 5 minutes until the download completes.

INSTALLATION STEPS

Step 1: As soon as you open the downloaded exe file, the install wizard will launch. Click on Run to start the installation process. You will get the first window as the welcome window. Check the box for license terms agreement and click on Install. This will start the install process.

Step 2: The next step will take a few minutes to install important files. The progress bar will indicate the progress in green.

Step 3: The next three steps are regarding the activation process. In this window, you have to choose from two options; Start trial now and Activate. The Start trial now option is the free 14-day trial option whereas, the Activate option is for installing the paid version in which you need to enter a product key.

Step 4: Moving forward, it will ask you to fill up a registration form. It contains fields like Name, E-mail, Organization, Job title, etc. Fill the right details and click on Register.

Step 5: Once the activation is complete, go to your desktop and click on the Tableau icon to open Tableau.

Step 6: Tableau Desktop welcome page will open. On its left, you can find a range of data source options you can connect to. And on its right panel, you can explore the training and resources option. To open a new workbook, click on the Tableau star-like icon given on the upper left corner.

DESCRIPTION OF THE TABLEAU WORKSPACE :

Menu Bar: It is the topmost bar; it contains File, Data, Worksheet, Dashboard, Story, Analysis, Map, Format, Server, Window, and Help. These options include features such as data exporting, file saving, etc.

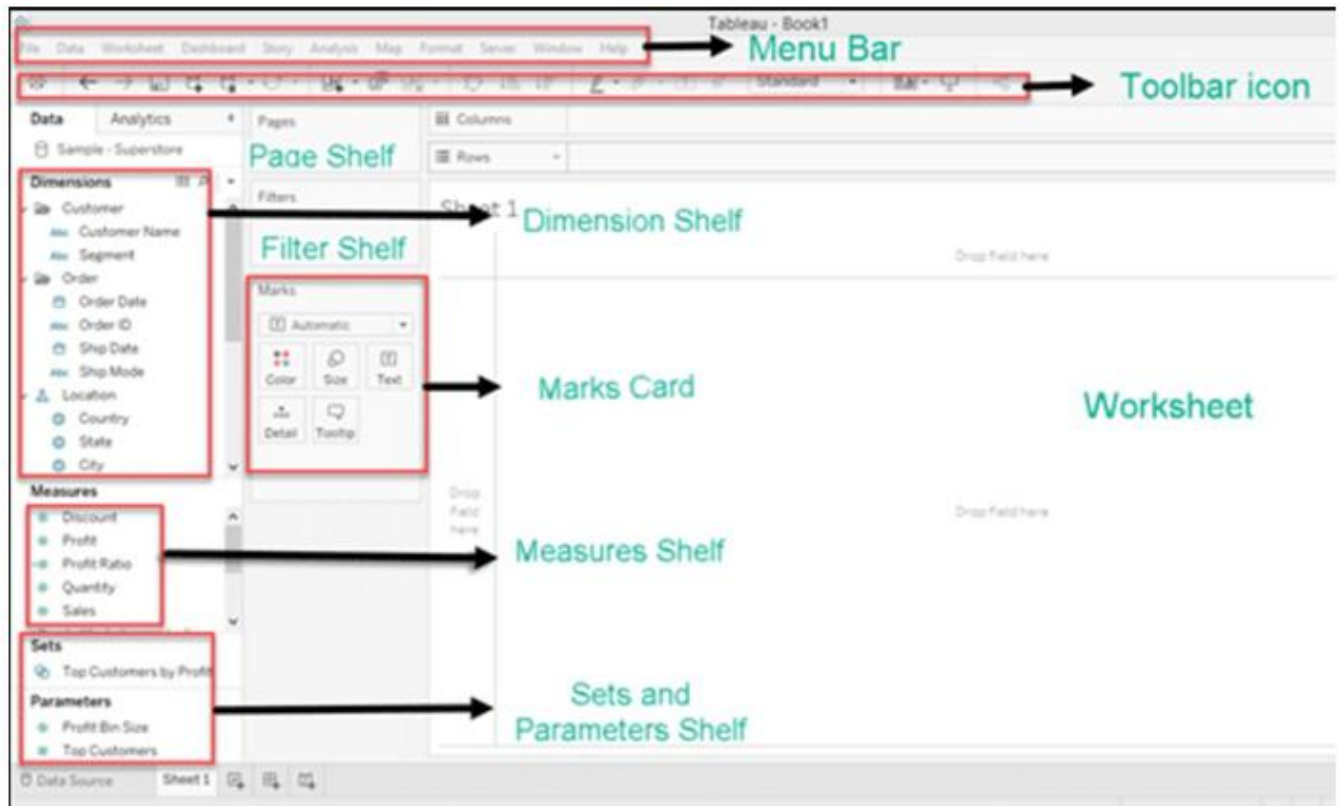
Toolbar: The tab present just below the Menu Bar is the Toolbar. It is basically used for editing the workbook and consists of various options such as view, undo, redo, slideshow, text edit, etc.

Dimension Shelf: The dimensions involved in the data sources can be viewed in the dimension shelf.

Marks Card: The options in the marks card are used to design the visualizations in the Tableau workspace.

Sets and Parameters Shelf: All user-defined sets and parameters can be viewed on this shelf. The existing sets and parameters can also be edited with the help of the options.

Tableau Repository: Tableau Repository is usually located in the file path C:\Users\User\Documents\My Tableau Repository. The repository is used to store all Tableau files; the repository is segregated into various folders such as Bookmarks, Connectors, Logs, Data Sources, etc.



RESULT :

Thus, Installation of Tableau desktop is successfully completed to perform business analytics.

Ex no : 2

Date :

COLLECTING CLEANING AND CONNECTING TO DATA**AIM**

To perform data collection and cleaning operations in tableau prep builder.

PROCEDURE

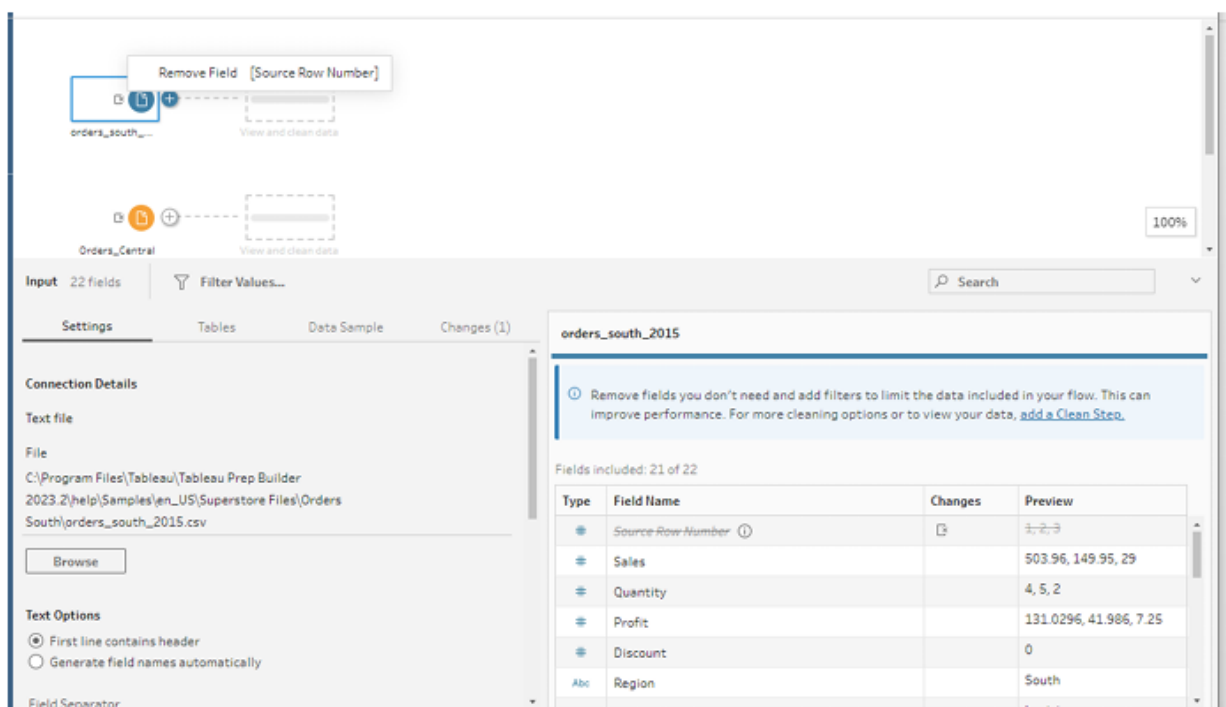
Step 1: Download and install the tableau pre-processing builder software 2023.2

Step 2: Run the tableau prep builder, Click on connect to data with text file, as we will connect .csv files as a text file.

Step 3: Go to this path “ C:\Program Files\Tableau\Tableau Prep Builder 2023.2\help\Samples\en_US\Superstore Files\Orders South\orders_south_2015.csv” and connect the file.

Step 4: Similarly, Connect the below two csv for preprocessing

- C:\Program Files\Tableau\Tableau Prep Builder 2023.2\help\Samples\en_US\Superstore Files\Orders_Central.csv
- C:\Program Files\Tableau\Tableau Prep Builder 2023.2\help\Samples\en_US\Superstore Files\Orders_Central.csv



Step 5: Click each file in the flow plane and review all the columns in the input pane of the particular table.

Step 6: You can observe that the Order date and ship date is different in all three tables, so make it same in all the tables.

Step 7: Create a clean step in Orders_South table, then click calculated field, Choose Field name as: Order Date And formula as `MAKEDATE([Order Year],[Order Month],[Order Day])`

Step 8: Do the same for Orders_Central table.

Step 9: Create a union and all the three tables and save it as a tableau file.

Step 10: Open the saved file in tableau desktop

OUTPUT:**RESULT**

Thus, to perform data collection and cleaning operations in tableau prep builder has been completed successfully.

Ex no : 3	ETL PROCESS FOR A SELECTED DATA SOURCE
Date :	

AIM:

To perform Extract Transform and Load (ETL) operations for a selected data source.

PROCEDURE :

Step 1: Download and install the tableau-desktop 2023.2.1.

Step 2: Download and Install MySQL Server (Windows (x86, 64-bit), MSI Installer) from

<https://dev.mysql.com/downloads/mysql/>

Step 3: Download and Install MySQL Workbench (Windows (x86, 64-bit), MSI Installer) from

<https://dev.mysql.com/downloads/workbench/>

Step 4: Download and Install MySQL connector (Windows (x86, 64-bit), MSI Installer) from

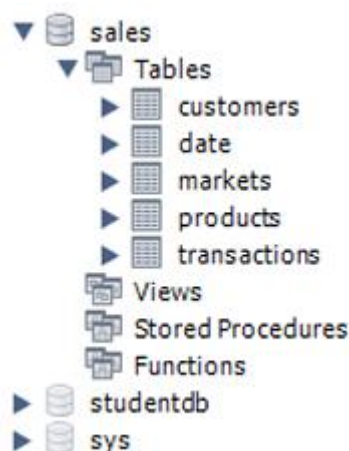
<https://dev.mysql.com/downloads/connector/odbc/>

Step 5: Download the SQL DB which consists of customers, date, markets, products and transactions tables of sales DB from

https://github.com/codebasics/DataAnalysisProjects/blob/master/2_SalesInsightsTableau/db_dump.sql

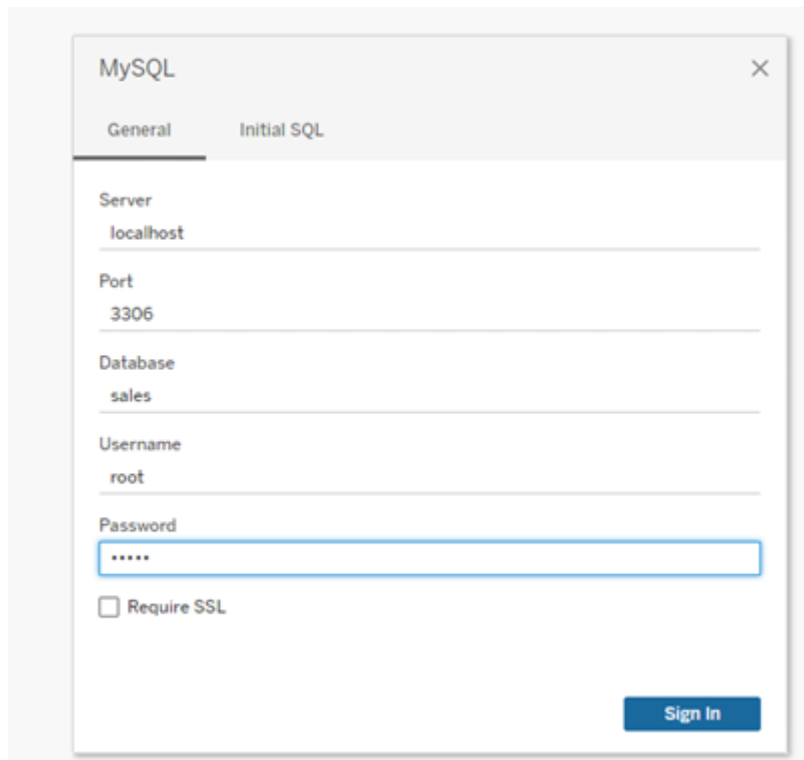
Step 6: Create mysql sales schema and import the downloaded db_dump.sql to sales DB.

Step 7: You can find all the sales tables will be imported in the sales DB.



Step 8: Keep the Server on in MySQL and open the tableau desktop.

Step 9: After opening tableau desktop, connect to MySQL Server with sales DB by specifying Username and password as shown below.



Step 10: After connecting, you can visualise all tables in the tableau, then drag the tables to form a star schema by specifying its relationships(keys), with fact and dimension table scenario as shown below in output.

Step 11: After extracting and connecting all the table data, perform transforming and loading process.

Step 12: Click on transaction table and inspect sales amount, we could able to find the sales amount is '-1' for a data, so remove it by applying the filter, data->edit data source filters. Once filter is applied we can observe that '-1' sales amount row will be deleted (As the sales amount would never be -1).

Step 13: Similarly, Inspect all the columns and transform the parameters like currency and sales amount and make necessary changes.

OUTPUT:

transactions+ (sales)

Connection: ☒ Live ☐ Extract Filters: 0 | A

transaction... — markets 17 rows

How do relationships differ from joins? [Learn more](#)

transactions Operator markets

Abc Market Code = Abc Markets Code

+ Add more fields

> Performance Options

Abc markets Markets Code	Abc markets Markets Name	Abc markets Zone
Mark001	Chennai	South
Mark002	Mumbai	Central
Mark003	Ahmedabad	North
Mark004	Delhi NCR	North
Mark005	Kanpur	North
Mark006	Bengaluru	South
Mark007	Bhopal	Central

RESULT :

Thus, Extract Transform and Load (ETL) operations for a selected data source have been completed successfully.

Ex no : 4	CREATE CHARTS LIKE BUBBLE, BAR AND MAP USING TABLEAU
Date :	

AIM:

To create charts like Bubble, Bar and Map using Tableau for a selected data source.

PROCEDURE :

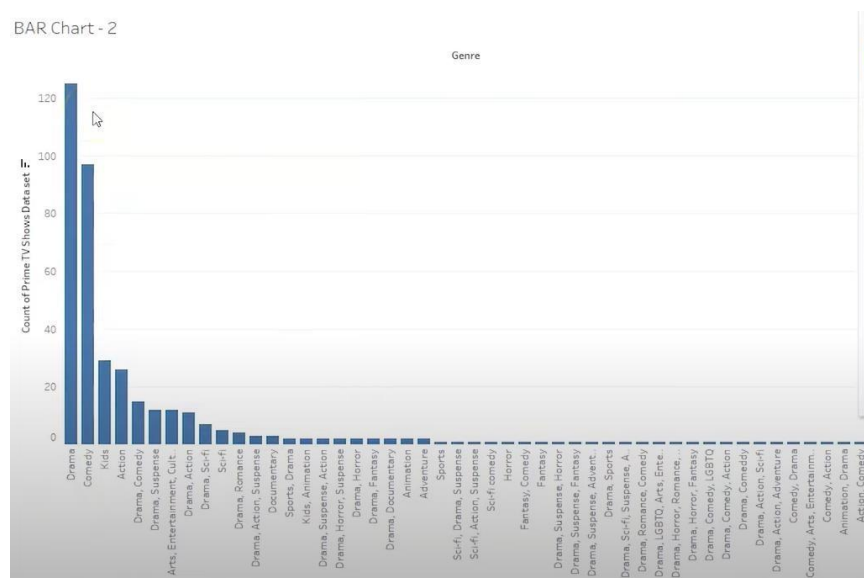
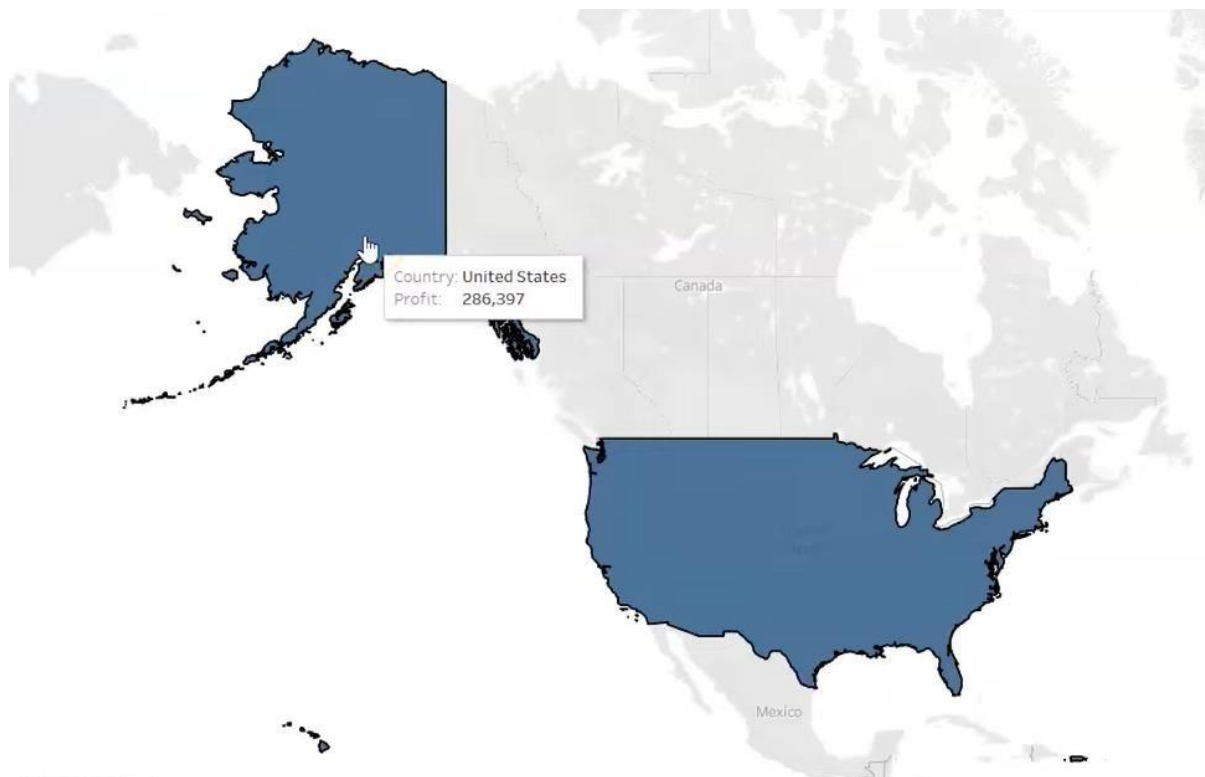
Step 1: Download amazon prime tv dataset from

<https://www.kaggle.com/datasets/shivamb/amazon-prime-movies-and-tv-shows/>

Step 2: Import the dataset into Tableau Desktop.

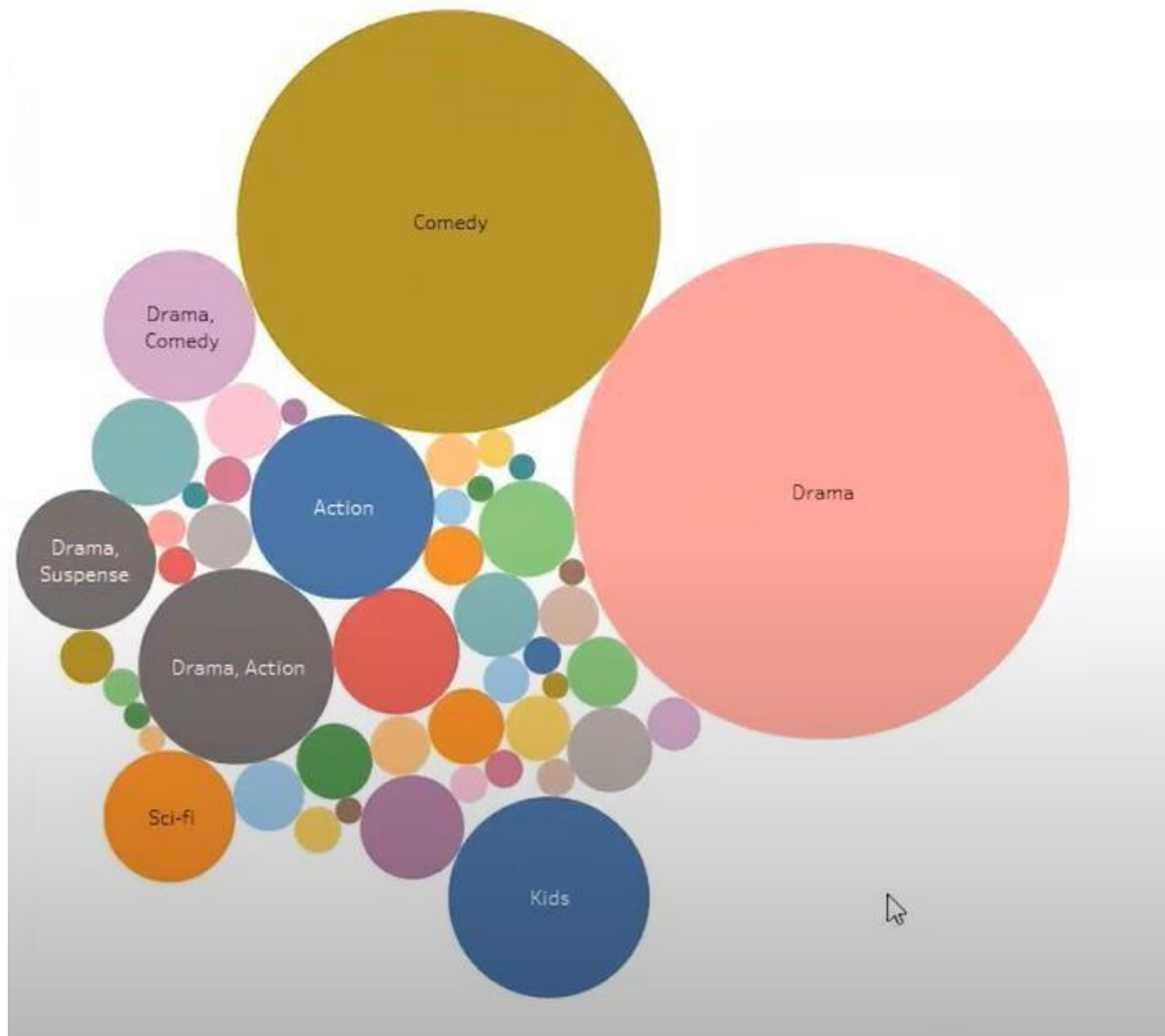
Step 3: Drag the attributes to the row or column space in tableau for creating the charts.

Step 4: In right side chart pane, choose the type of chart to visualize the selected attributes.

OUTPUT:**Bar Chart :****Map Chart :**

Bubble Chart :

Buble

**RESULT:**

Thus, charts like Bubble, Bar and Map has been built successfully using tableau.

Ex no : 5	CREATE YOUR STORY FROM THE CHARTS WITH VALID REASONS
Date :	

AIM:

To create story from tableau for a selected data source.

ALGORITHM :

Step 1: Import the super store data to tableau desktop from C:\Program Files\Tableau\Tableau Prep Builder 2023.2\help\Samples\en_US .

Step 2: Create the charts that are needed by choosing attributes.

Step 3: Connect to the tableau server using your tableau login.

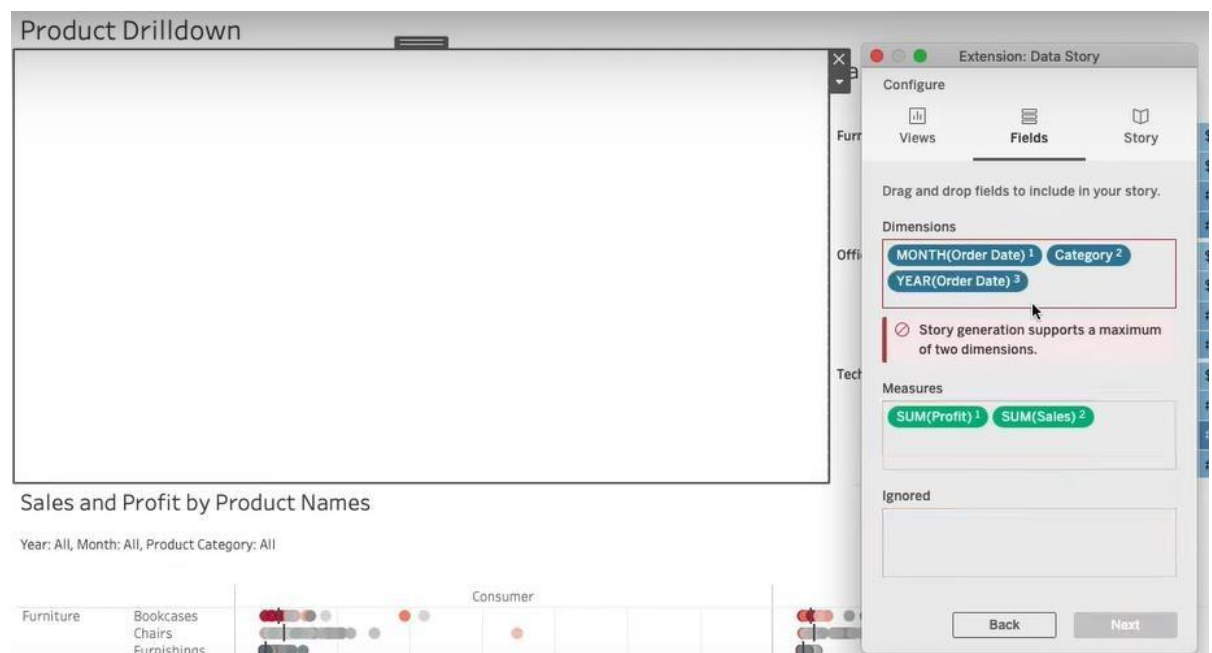
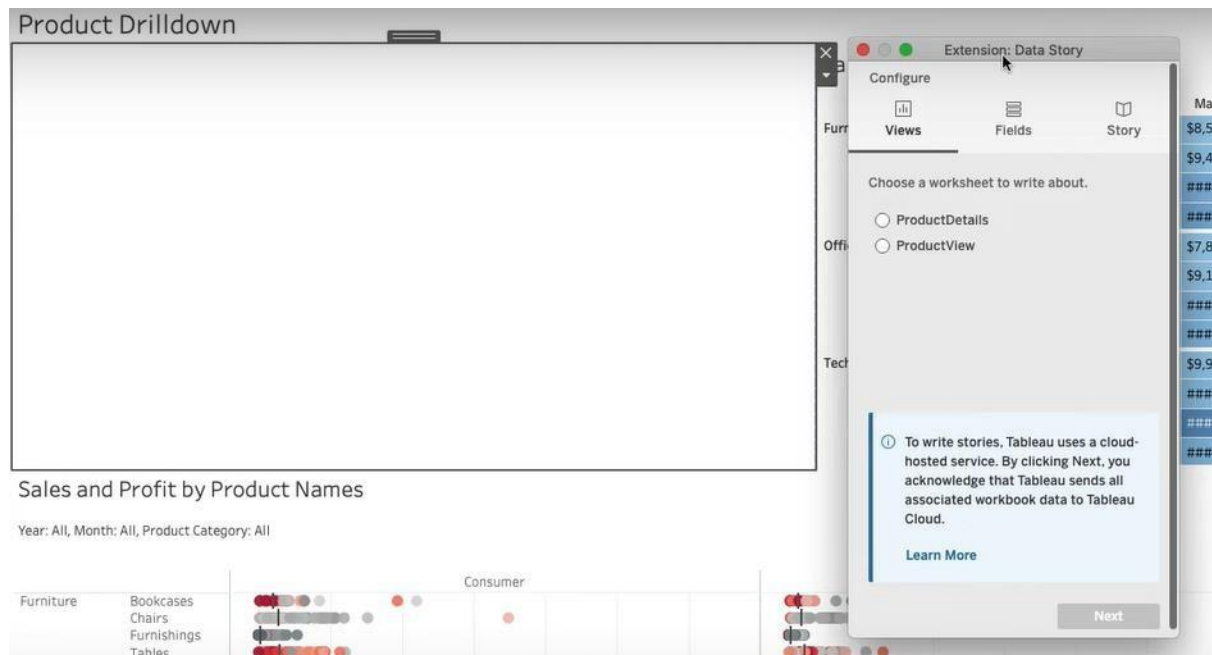
Step 4: In left side pane, double click on the data story icon and configure dialog box will Appear.

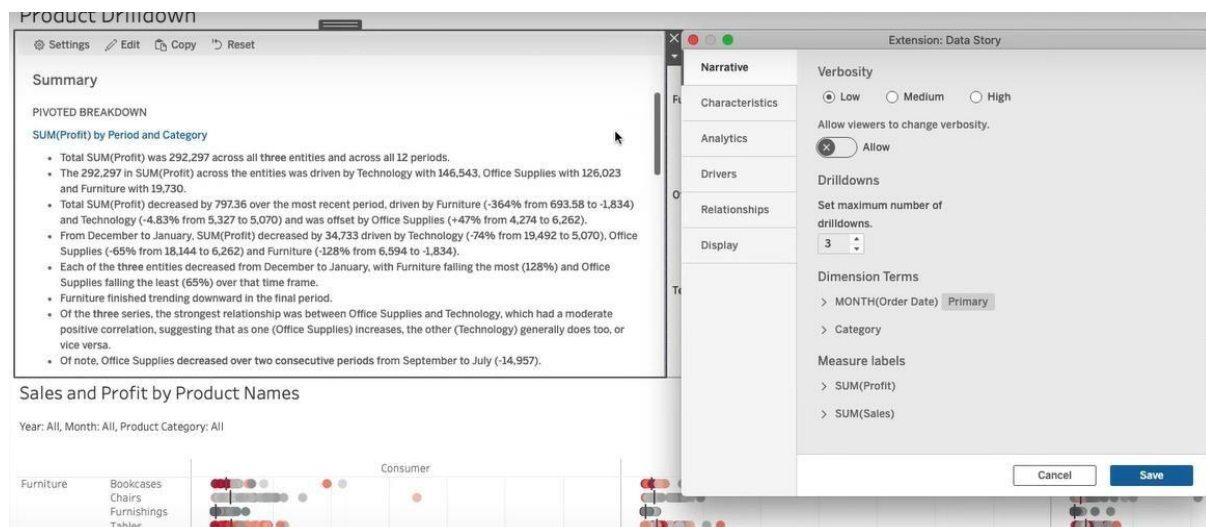
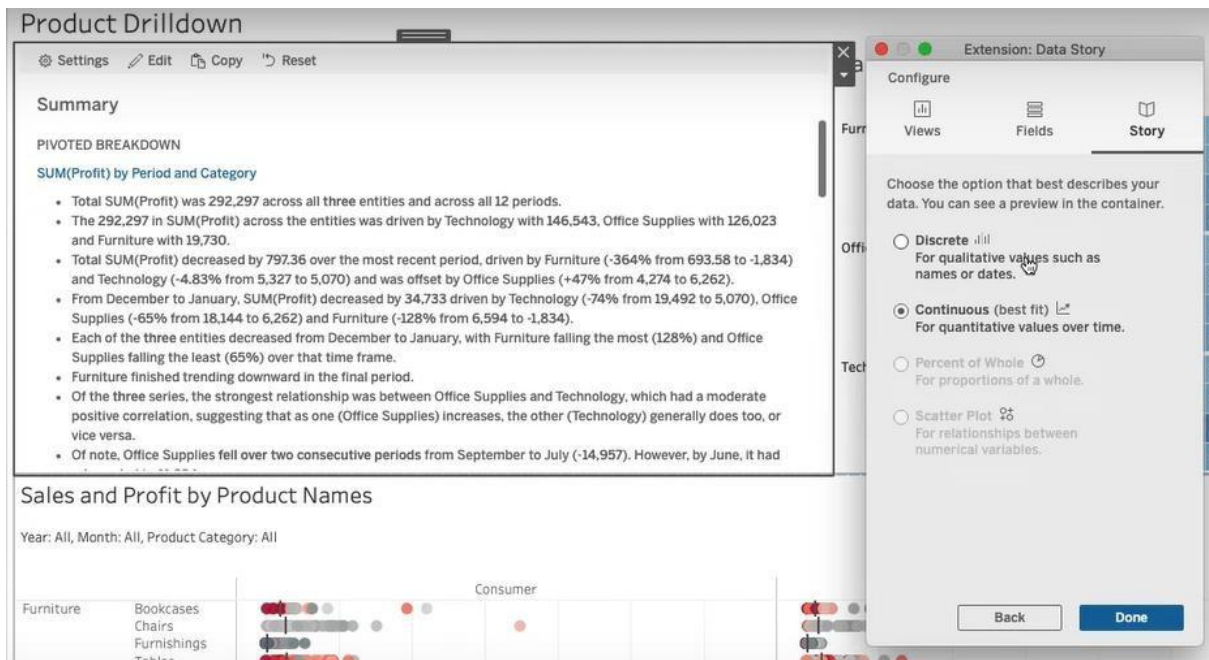
Step 5: In views, choose the work sheet for which the story has to be created.

Step 6: In fields, select the fields on which the story has to flow and create a measure on basis the story has to flow (Eg: AVG(field_name)).

Step 7: In Story, Select the type of story to generate. Most probably select continues stories type and give done. Story will appear on left side.

Step 8: We can change the characteristics, Analytics and Display of your choice from the settings in the story.

OUTPUT :



RESULT:

Thus story from tableau for a selected data source has been built successfully.

Ex no : 6	BUILD A MAP VIEW USING TABLEAU (EXPLORE YOUR DATA GEOGRAPHICALLY).
Date :	

AIM:

To create a map view using tableau for a selected data source.

PROCEDURE

Step 1: Import the super store data to tableau desktop from

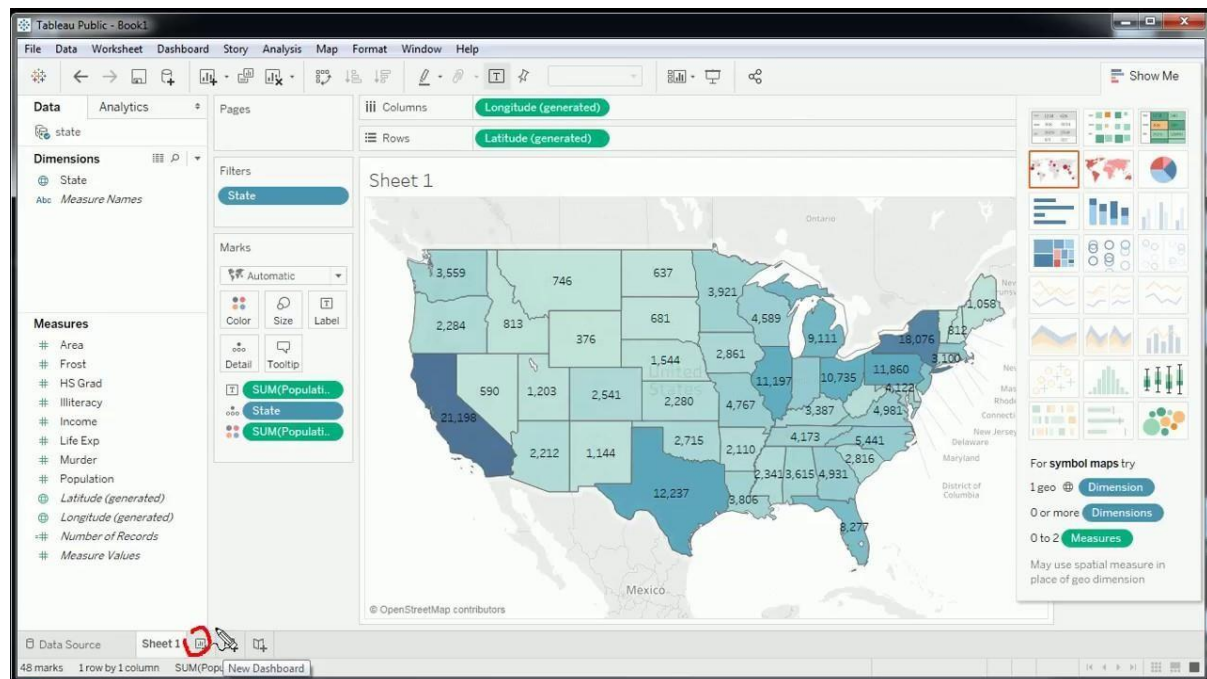
C:\Program Files\Tableau\Tableau Prep Builder 2023.2\help\Samples\en_US

Step 2: Select the attributes for which the map view is needed (Location Attribute is mandatory)

Step 3: If live data is needed then connect to the tableau server using your tableau login

Step 4: In right side pane, double click on the geographical chart for map view of the data.

Step 5: Map view will be created and displayed in the workspace .

OUTPUT:**RESULT:**

Thus a map view using tableau for a selected data source has been built successfully.

Ex no : 7	COLLECT APPROPRIATE DATA, PERFORM ETL PROCESS AND DEVELOP SCORECARD USING AIR FLOW AND TABLEAU.
Date :	

AIM:

To collect relevant data, perform the ETL (Extract, Transform, Load) process, and develop a Scorecard using Tableau for effective data visualization and analysis_.

PROCEDURE:**Step 1: Data Collection**

1. Identify a dataset relevant to the business domain (e.g., healthcare, sales, or finance).
2. Download the dataset in CSV, Excel, or connect to an SQL database.
3. Ensure data completeness and quality.

Step 2: Perform ETL Process**1. Extract:**

- Use **Tableau Prep Builder** or other ETL tools to load the dataset.
- Connect to multiple data sources if required.

2. Transform:

- Clean data by handling missing values, duplicate entries, and inconsistent formatting.
- Perform calculations and create new columns (e.g., revenue per customer, total transactions).
- Apply filters to remove irrelevant data.
- Standardize date formats and currency units.

3. Load:

- Save the cleaned data and load it into **Tableau Desktop**.
- Verify the correctness of transformations.

Step 3: Develop Scorecard in Tableau

1. Connect to the Cleaned Data:

- Open Tableau Desktop.
- Load the transformed dataset.

2. Create Key Performance Indicators (KPIs):

- Define critical KPIs (e.g., revenue growth, sales performance, patient satisfaction).
- Use **calculated fields** in Tableau to derive KPIs.

3. Design Visualizations:

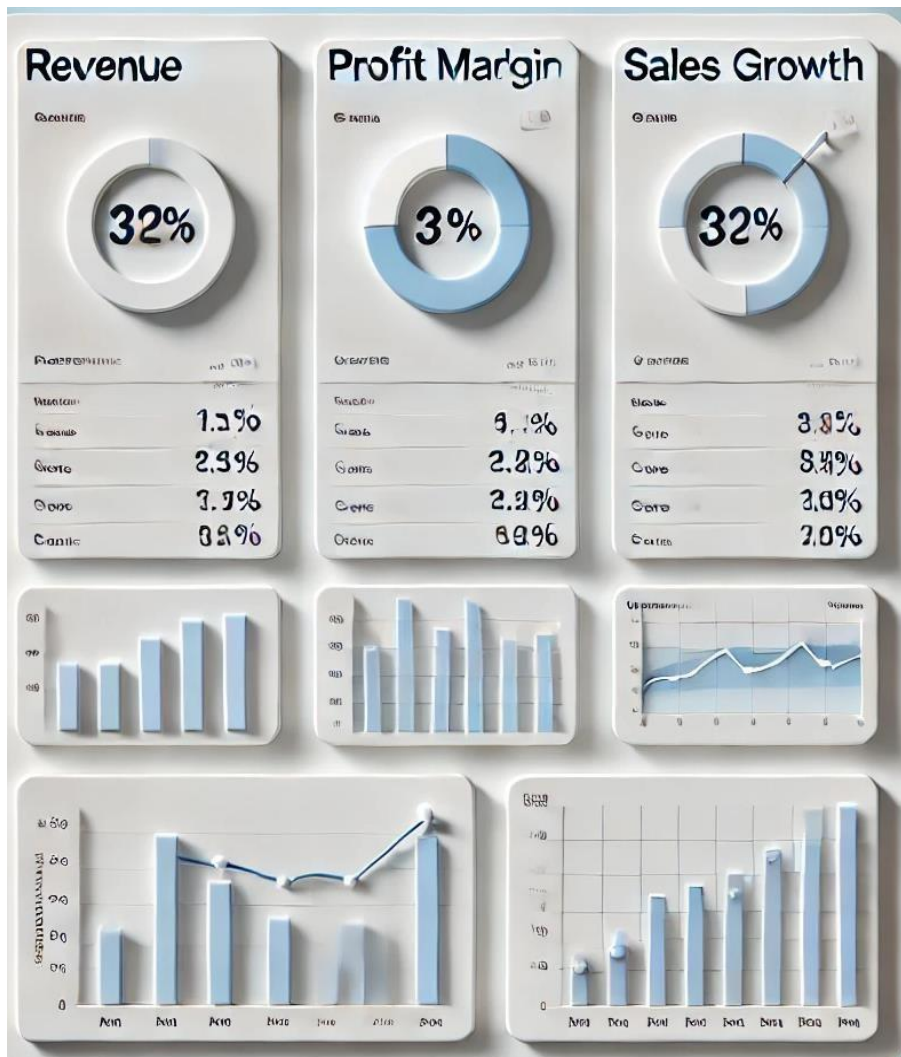
- Use **Bar Charts, Line Graphs, KPI Cards, and Gauge Charts** to display performance indicators.
- Apply **conditional formatting** to highlight KPIs (e.g., red for low, green for high performance).
- Use **filters** and **parameters** to allow dynamic user interactions.

4. Build the Scorecard:

- Create a **dashboard** in Tableau and arrange KPI visualizations.
- Add interactivity using **filters, drop-down menus, and tooltips**.
- Format the layout for clarity and ease of interpretation.

5. Publish and Share:

- Publish the Scorecard to **Tableau Server or Tableau Public**.
- Share with stakeholders for business insights .



RESULT:

Thus, the ETL process was successfully performed, and a Scorecard was developed using Tableau to visualize key metrics effectively.

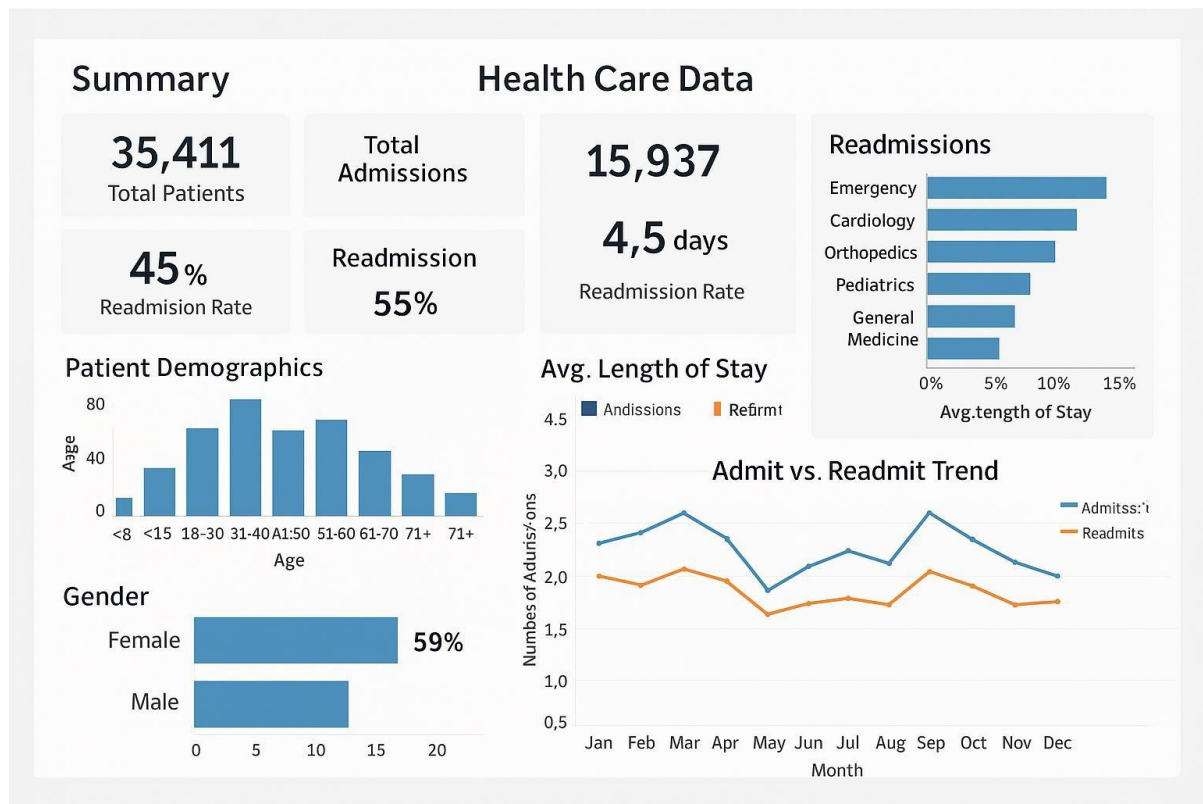
Ex : 8	COLLECT APPROPRIATE DATA, PERFORM ETL PROCESS AND DEVELOP DASHBOARD FOR HEALTH CARE USING AIR FLOW AND TABLEAU.
Date:	

AIM:

To perform ETL process and develop a dashboard for a selected health care data Source .

PROCEDURE

- Step 1: Perform all the ETL process by inspecting each and every attributes.
- Step 2: Create the charts that are needed by choosing attributes in a separate sheet.
- Step 3: Connect to the tableau server using your tableau login if needed
- Step 4: Choose the background image (png format) for dashboard with separate panes
- Step 5: Import the image into tableau and drag the workout tableau sheets into the panes of the imported image.

OUTPUT:**RESULT:**

Thus ETL process and development of a dashboard for a selected health care data source has been completed successfully.